

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	examine properties of natural and made materials and explain how those properties influence their use
KEY IDEA	Materials are chosen for observable and testable properties such as strength, flexibility, absorbency, transparency, waterproofing and heat conduction - not simply because they are natural or made.
WORKED MODEL	Clear glass suits a window because it is transparent and rigid, but its brittleness may make another material safer for some uses.

SUCCESS CRITERIA - I CAN

[] use key vocabulary [] classify or model accurately [] explain using evidence [] correct a misconception

A. BUILD THE SCIENCE IDEA

1. Define and use these terms: property and natural material.

2. Match examples to: property, natural material, made material.

3. Annotate the worked model with two observations and one scientific explanation.

B. APPLY AND EXPLAIN

4. Sort six examples from this context: cotton, wool, timber, steel, glass, plastic and ceramic

5. Explain two classifications using observable evidence, not appearance alone.

6. Identify the changed condition, the effect and one condition that should stay the same.

C. INDEPENDENT CHECK

7. Draw or construct a labelled scientific model that shows the relationship.

8. Write a three-sentence explanation using because, therefore and evidence.

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D. APPLY SCIENCE IN CONTEXT

9. Case study: A designer compares cotton, coated fabric and plastic for a reusable lunch bag.

10. Record the important observations in a two-column evidence table.

11. Write a claim-evidence-reasoning conclusion and state one limitation.

E. REASON, EVALUATE AND TRANSFER

12. Find and correct the inaccurate statement in the case study; explain why it is inaccurate.

13. Apply the same science idea to a different everyday situation and predict what would happen.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can examine properties of natural and made materials and explain how those properties influence their use.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Property is a characteristic; natural comes from living/Earth sources; made is manufactured or processed.
2. Accept classifications noting many materials are processed.
3. Transparency and rigidity support glass use; brittleness is a limitation.
4. Sort with explanation where origin is complex.
5. Evidence can include absorbency, strength, flexibility or waterproofing.
6. Changed: material; measured: leak/strength/temperature; controls include size and test conditions.
- 7-8. Model links property -> advantage/limitation -> use.
9. Waterproofing, insulation, durability, cleanability and environmental impact matter.
10. Table should compare the same properties fairly.
11. Recommend with evidence and acknowledge limited tests.
12. Natural does not automatically mean safer or better.
13. Accept clothing, shelter, packaging or tool example.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Natural and made are origins, not quality rankings.

One useful property may come with a limitation; material selection involves trade-offs.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.